

Take-Home Assignment: Alternative Data Research — Datadog (NASDAQ: DDOG)

Background

Datadog (DDOG) is a premier cloud observability and security platform, generating revenue primarily through usage-based and subscription models that are closely correlated with customers' cloud workloads. As part of our interview process, we invite you to conduct a concise research project exploring the application of high-frequency and alternative data to track and anticipate DDOG's business performance in advance of quarterly earnings releases.

Objective

Your goal is to identify and critically evaluate high-frequency and alternative datasets capable of serving as leading indicators for Datadog's key financial and operational metrics. These may include, but are not limited to: quarterly revenue and growth rates, billings or remaining performance obligations (RPO), net revenue retention (NRR), and customer expansion trends (e.g., number of customers with ARR $\geq$ \$100k).

Assignment Components

1. Data Selection and Rationale

Propose **3–5** high-frequency or alternative data sources that you believe offer the most predictive value for DDOG's performance. For each dataset, please address:

- **Definition & Origin:** What the data measures and its source.
- **Economic Rationale:** A logical explanation of why this data should lead or track DDOG's revenue model. Mere plausibility is insufficient—support your argument with concrete evidence, such as company disclosures (e.g., commentary on usage-based pricing, customer mix, cloud partnerships), management guidance, industry research, academic or practitioner studies, or your own preliminary empirical observations.
- **Data Characteristics:** Frequency, latency, historical coverage, and breadth.
- **Accessibility & Cost:** Indicate whether the data is freely available, scrapable, or requires vendor procurement.
- **Limitations & Mitigations:** Identify key biases or constraints and propose strategies to address them.

2. Statistical Analysis and Predictive Evidence

Leveraging publicly accessible data wherever feasible, test **at least two** of your proposed signals against DDOG's reported results over the past **8–12+ quarters**. Employ appropriate statistical techniques to demonstrate and refine their predictive accuracy. Your analysis should include:

- **Lead–Lag Analysis:** Evidence that your signals precede, rather than simply coincide with, reported KPIs.
- **Predictive Modeling:** A model linking your signals to quarterly revenue or revenue growth—examples include linear or regularized regression, a time-series framework, or a nowcasting approach.
- **Out-of-Sample Validation:** Walk-forward or rolling backtests that report error metrics (e.g., MAPE, RMSE) and directional hit rates.

Every predictive claim must be substantiated by this evidence. Be explicit about what your data supports and what it does not. Given the limited quarterly history, we highly value rigorous methodology and honest interpretation over complexity: a well-validated simple model is preferred over an overfitted, intricate one.

3. Dashboard Design

Design a preliminary dashboard that an analyst would use to monitor these signals ahead of earnings. A working prototype (e.g., using Streamlit, Tableau, Power BI, Excel) or a well-annotated mockup (e.g., Figma, slides) is equally acceptable. The design should convey:

- **Key Visuals & Metrics:** Core charts and indicators, along with their update frequencies.
- **Signal Synthesis:** How individual signals are combined to produce a revenue estimate or a directional call for the quarter.
- **Interpretation Guide:** What "tracking ahead" versus "tracking behind" would look like in the visualized data.

Deliverables

1. **Written Report:** A concise report (max 5 pages or 10 slides) covering data selection, rationale, methodology, findings, and limitations.
2. **Dashboard Prototype/Mockup:** A prototype or design mockup with brief annotations.
3. **Code (Optional):** Any supporting code or notebooks used for data collection and analysis.

Important Notes

- **Data Compliance:** Use only publicly available or legally accessible data. Do not use material non-public information (MNPI), and ensure all data collection adheres to relevant websites' terms of service.
- **Open-Ended Approach:** There is no single "right answer." We are most interested in your analytical reasoning, creativity, and how you structure your problem-solving process.